LUCAS MOUGEOT

Product Engineer – Data & LLM Systems | Deterministic AI Infrastructure

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SUMMARY

Systems engineer specializing in deterministic data infrastructure and production LLM systems. I design observable, schema-driven platforms that transform unstructured inputs into reliable, auditable decision systems. Experienced building Python backend services, PostgreSQL data models, and LLM orchestration frameworks within containerized multi-service architectures. Strong foundation in physical systems and AI integration, with a focus on latency control, hallucination mitigation, and reproducible production deployments.

EXPERIENCE

Integration Engineer — EZ Systems (Full-Time) Feb 2025 – Aug 2025

- Designed and implemented a real-time ADA compliance auditor combining computer vision, LiDAR-derived measurements, and parsed regulatory rules into a structured, decision-ready system.
- Architected a unified QA and deployment framework orchestrating AI inference, sensor calibration, and CI/CD pipelines; defined mechanisms for model drift mitigation, versioned inference tracking, drift detection, and reproducible deployment across physical and AI-integrated systems.
- Developed robotic scan-path generation software from customer-provided STEP models, translating geometric constraints into executable motion plans.
- Led engineering for a self-driving lawn mower pilot program operating in solar-field environments.
- Authored technical strategy documents including federal grant proposals on explainable AI and internal AI integration standards.

Founding Systems Engineer — Stealth Platform 2024 – Present

- Architect and own a production-grade AI-assisted compliance and decision platform (Python, TypeScript, PostgreSQL) spanning structured intake, rule evaluation, attribution, and auditable decision workflows.
- Managed end-to-end deployment cycles including configuration versioning, rollback strategies, monitoring dashboards, and structured logging to support production stability and controlled iteration.
- Designed a deterministic ingestion engine in Python that compiles arbitrary repositories into immutable structural and dependency graphs via AST-based analysis, enabling reproducible context assembly for LLM-driven coding and classification.
- Built LLM orchestration and observability tooling including prompt versioning, structured input/output logging (latency, token cost, metadata), evaluation pipelines, and failure-mode analysis to iteratively reduce hallucination and improve response reliability in production workflows.
- Developed controlled tool-execution patterns enabling LLMs to call deterministic backend services (database queries, validation services, rule engines) while enforcing structured schemas and validation layers to prevent unsafe or invalid state transitions.
- Optimized context construction and retrieval paths to balance latency, token cost, and response quality across LLM workflows.
- Implemented hybrid retrieval systems (vector + full-text) with citation anchoring and validation layers to enforce traceability and reduce hallucination in production workflows.
- Defined data schemas and lineage boundaries supporting versioned workflows, reproducibility guarantees, and cross-service consistency.
- Enforced schema-first system boundaries isolating probabilistic LLM components behind deterministic validation layers, enabling replayable state transitions, cost tracking, and reproducible production behavior.
- Deployed multi-service Docker architecture (PostgreSQL, Redis, BullMQ) with regression, fuzz, and concurrency testing to guarantee stable behavior under parallel workloads.

Graduate Researcher — Materials Science (NASA-Funded Capillary Devices) 2023 – 2024

- Designed and fabricated porous ceramic devices enabling passive capillary transport in microgravity environments.
- Built MATLAB computer-vision pipelines for pore segmentation, morphology quantification, and statistical ranking of candidate formulations.

- Authored experimental SOPs, DOEs, and failure-mode documentation; delivered a top formulation achieving ~3× NASA baseline performance in bench tests.

Undergraduate Researcher — Computational Geometry (NSF-Funded) 2020 – 2022

- Implemented bounded-degree planar geometric spanners in modern C++ using CGAL under strict stretch-factor and planarity constraints.
- Designed locality-aware spatial algorithms for dense and heterogeneous point datasets.
- Contributed to peer-reviewed ACM Journal of Experimental Algorithmics publication.

SELECTED PROJECTS

Automated Lithium-Ion Battery Pack Assembler Senior Design Capstone

- Designed and implemented an automated lithium-ion battery pack assembly system using an AutomationDirect Do-More PLC.
- Developed ladder logic coordinating motion sequencing, interlocks, fault handling, and safe state transitions.
- Integrated an RP Series robotic dispenser via RS-232 serial communication to initiate external job file execution.
- Implemented PWM-based servo motor control for precise positioning and timing.
- Delivered a deterministic, repeatable automation pipeline with full I/O mapping and bring-up procedures prior to generative AI tools.

EDUCATION

University of North Florida

M.S. Materials Science & Engineering (Thesis-Based) — GPA: 4.00 / 4.00

- Thesis: Development and Characterization of Ceramic Devices for Capillary Action in Microgravity for the Purpose of Growing Vegetables in Space (NASA-Funded)

B.S. Electrical Engineering — GPA: 3.93 / 4.00

B.A. Interdisciplinary Studies: Morphology of Futurism — GPA: 4.00 / 4.00

Concentrations: Mechanical Engineering, Materials Science, Chemistry, Physics, Computing, Mathematics, Mandarin Chinese

PUBLICATIONS & PRESENTATIONS

- Bounded-Degree Plane Geometric Spanners in Practice — ACM Journal of Experimental Algorithmics (Peer-Reviewed)
- Highly Sensitive and Stable pH Sensor Electrodes of TiN Fabricated Using HiPIMS with Kick — AVS 69 Poster, 2023
- Interactive Tool for Bounded-Degree Plane Geometric Spanners — SoCG Media Exposition, 2021

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, MATLAB

Data & AI Systems: Agentic LLM workflows, LLM orchestration, hybrid RAG (vector + full-text), embeddings, prompt/version management, structured evaluation pipelines, LLM observability (latency, cost, metadata), deterministic ingestion engines, graph-based context assembly, lineage tracking

Infrastructure: Docker, PostgreSQL, Redis, BullMQ, GitHub Actions, AWS-based deployments

Backend & APIs: FastAPI-style services, background workers, schema design, access control, concurrency-safe systems

Hardware Context: Electrical engineering, LiDAR integration, robotics motion planning, sensor calibration